

- 6 (a) (i) magnitude of induced e.m.f. in a conductor is directly proportional to the rate of change of magnetic flux linkage. B1
- (ii) Recognise that the value is given by area under the graph from $t = 0$ to $t = 0.025$ s C1
- Show substitution using trapezium rule or equivalent. (When integration is used, the function used must be clear) A1
- Acceptable range 0.38 ± 0.04 Wb
- (b) (i) $0.38 = B \times 0.090 \times 8$ C1
 $B = 0.53$ Wb A1
- (ii) $\omega = \frac{2\pi}{0.10} = 20\pi \text{ rad s}^{-1}$ B1
- (c) Straight line through origin B1
- (d) (i) $V_0 = I_0 R$, so $I_0 = \frac{24}{12} = 2.0$ A C1
 $I_{\text{rms}} = \frac{2.0}{\sqrt{2}} = 1.4$ A A1
 or equivalent method
- (ii) *expression needs to be updated, quite misleading.
 mean power = $\frac{P_0}{2} = \frac{I_0 V_0}{2} = \frac{24 \times 2.0}{2} = 24$ W B1
 or equivalent method